

UNIT 01

CHEMISTRY OF REPRESENTATIVE ELEMENTS



Elements in the periodic table from group I-A to VIII-A are called representative elements. They are found in excess amount in nature that's why they're called as representative elements.

S-block Elements:

Elements of group I-A and II-A are included in s-block elements. The general valence shell electronic configuration of s-block elements can be given by ns^{1-2} .

- Elements of group I-A are also called “Alkali Metals” and the general valence shell electronic configuration of this group is ns^1 .
- Elements of group II-A are also called “Alkaline Earth Metals” and the general valence shell electronic configuration of this group is ns^2 .

GIA



Lithium (Li)



Sodium (Na)



Potassium (K)

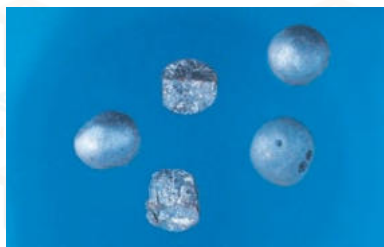


Rubidium (Rb)



Cesium (Cs)

Gila



Beryllium (Be)



Magnesium (Mg)



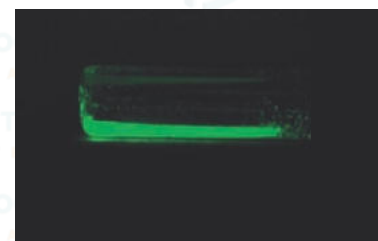
Calcium (Ca)



Strontium (Sr)



Barium (Ba)



Radium (Ra)

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1A	Cesium (Cs)																8A					
2A																	3A	4A	5A	6A	7A	
Be																						
Mg																						
Ca																						
Sr																						
Ba																						



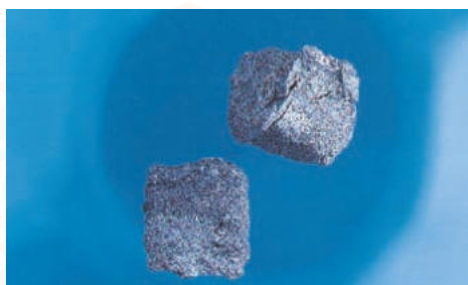
P-Block Elements:

The elements in which the valence electron enters in orbital is called p-block elements.

Elements of group III-A to VIII-A are included in p-block elements.

- The general valence shell electronic configuration of p-block elements can be given by ns^2, np^{1-6} . Elements of group III-A are also called "Boron Family" and the general valence shell electronic configuration of this group is ns^2, np^1 .
- Elements of group IV-A are also called "Carbon Family" and the general valence shell electronic configuration of this group is ns^2, np^2 .
- Elements of group V-A are also called "Nitrogen Family or Pnictogens" and the general valence shell electronic configuration of this group is ns^2, np^3 .
- Elements of group VI-A are also called "Oxygen Family or Chalcogens" and the general valence shell electronic configuration of this group is ns^2, np^4 .
- Elements of group VII-A are also called "Halogens" because they are salt formers and the general valence shell electronic configuration of this group is ns^2, np^5 .
- Elements of group VIII-A are also called "Noble gases or Inert gases" and the general valence shell electronic configuration of this group is ns^2, np^6 .

8A							
7A							
6A							
5A							
4A							
3A	B	Al	Ga	In	Tl		
2A							
1A							



Boron (B)



Aluminum (Al)



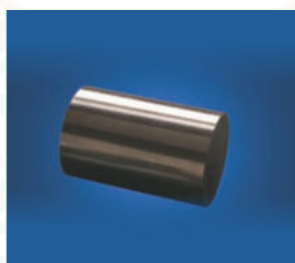
Gallium (Ga)



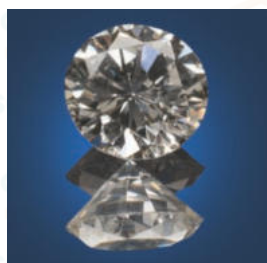
Indium (In)

G-III A

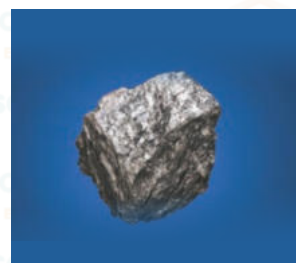
8A							
7A							
6A							
5A							
4A	C	Si	Ge	Sn	Pb		
3A							
2A							
1A							



Carbon (graphite)



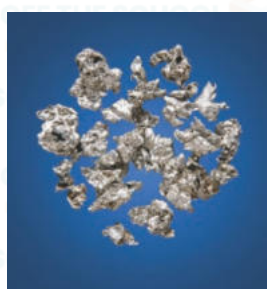
Carbon (diamond)



Silicon (Si)



Germanium (Ge)



Tin (Sn)



Lead (Pb)

G-IV A**CHEMISTRY OF REPRESENTATIVE ELEMENTS**

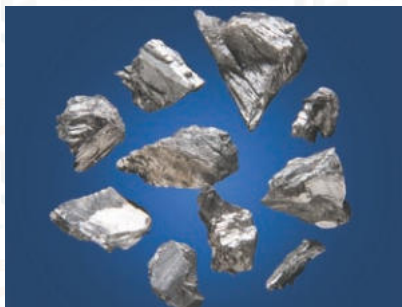
G-VA



White and red phosphorus (P)

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Arsenic (As)



Antimony (Sb)



Bismuth (Bi)

G-VIA

1A	2A	THE SCHOOL										3A	4A	5A	6A	7A	8A
																O	
																S	
																Se	
																Te	
																Po	



Sulfur (S_8)



Selenium (Se_8)



Tellurium (Te)

The Group 6A elements sulfur, selenium, and tellurium. Molecular oxygen is a colorless, odorless gas. Polonium (not shown) is radioactive.

1A	OFF THE SCHOOL																8A	
2A	OFF THE SCHOOL																3A 4A 5A 6A 7A	
		OFF THE SCHOOL																F
		OFF THE SCHOOL																Cl
		OFF THE SCHOOL																Br
		OFF THE SCHOOL																I
		OFF THE SCHOOL																At

G-VIA



[illegible]

Neon (Ne)

Argon (Ar)

Krypton (Kr)

Xenon (Xe)

All noble gases are colorless and odorless.

These pictures show the colors emitted by the gases from a discharge tube.

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The ground-state electron configurations of the elements. For simplicity, only the configurations of the outer electrons are shown.

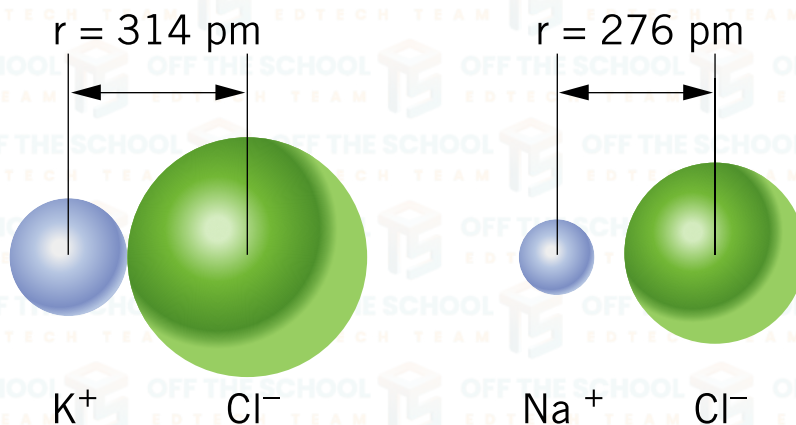


1. ATOMIC RADIUS:

Distance between the nucleus of an atom and its valence shell (i.e., the outermost shell that is occupied by one or more electrons), because we usually envision atoms as spheres with discrete boundaries.

According to Quantum mechanical model, this definition considers the fact that the electrons in an atom are not confined to a specific radius, but rather exist in a cloud of probability.

“The distance between the centers of two bonded atoms of any elements. Half of this distance is considered to be the radius of the atom.”

**General Group trend:**

The group trend of atomic radius for representative elements is that it increases from top to bottom. This is because the number of electron shells increases as you move down a group. Each electron shell is located farther away from the nucleus than the previous shell. This means that the electrons in the outermost shell are farther away from the nucleus than the electrons in the outermost shell of an atom in the same Group.

Exceptions:**Al & Ga (G-III A)**

The atomic radius is like the size of an atom. In simple terms, it's how far the outermost electrons are from the nucleus. When comparing Al (143pm) and Ga (135pm), we find that Al has a larger atomic radius, and one reason for this is something called "poor shielding of d orbitals."

2. IONIZATION POTENTIAL/ENTHALPY/ENERGY:

“The amount of energy required to remove an electron from a neutral, isolated and gaseous atom.”

OR

“The Ionization energy is the energy required to remove one electron from one mole of a ground state atom of the element in the gaseous state to form one mole of a gaseous uni positive (1+) ions.”

Explanation

In chemistry, ionization potential is a bit like that pulling energy. It's the amount of energy you need to take an electron away from an atom. When you do this, the atom turns into an ion because it now has an unequal number of electrons and protons.

So, the ionization potential is like a measure of how tightly those electrons are holding onto the nucleus of the atom. If it takes a lot of energy to pull an electron away that means the ionization potential is high, and the electrons are strongly attached. But if it doesn't take much energy, then the ionization potential is low, and the electrons aren't as tightly held.

General group trend:

When you go down a group, the ionization potential generally decreases.

This decrease occurs because, as you move down a group, you're adding more energy levels (shells) to the atom. These outermost electrons are farther from the nucleus and are shielded by the inner electrons. The increase in energy levels and the shielding effect make it easier to remove an electron from the outermost shell. The outer electrons feel less of the attractive force from the nucleus (Effective Nuclear Charge), so it takes less energy to remove them.

Nuclear charge (Z): is simply the number of protons in the nucleus of an atom. Effective nuclear charge (Z_{eff}) is the actual magnitude of positive charge that is "experienced" by an electron in the atom.

Shielding effect: the shielding of inner-shell electrons to the outer-most electrons is simply shielding or screening effect.

Exceptions:**1) Al (Aluminum) and Ga (Gallium) - Group IIIA:**

The general trend for ionization potential in Group IIIA is that it decreases as you move down the group due to increased atomic size.

However, there's an exception when you compare Al and Ga. Gallium has a lower ionization potential than aluminum.

The primary reason for this exception is the "poor shielding" effect of the d-orbitals in Gallium.

2) Tl (Thallium) and In (Indium) - Group IIIA:

In Group IIIA, the ionization potential generally decreases as you go down the group. Thallium, however, has a higher ionization potential than indium, which is an exception to the trend.

This exception occurs because thallium has its outermost electron in a relatively poor shielding 4f orbital, while indium has its outermost electron in a better shielding 5s orbital. The 4f orbital does not shield the outer electrons as effectively as the 5s orbital, making it easier to remove the outermost electron in indium, hence the lower ionization potential.

3) Sn (Tin) & Pb (Lead) - Group IVA:

The atomic size of tin and lead are very close, and hence, the charge density is more in case of lead, due to more charge. Due to higher charge density, the outer electrons are attracted more by the nucleus in lead and hence, the ionization energy is more in the case of lead.

Give reasons for the following:**• The ionization energy of Na is more than K.**

Ans. Sodium contains a smaller number of shells so that it is difficult to remove its electron due to strong electrostatic attraction. Hence, sodium has more ionization energy than potassium.

• The ionization energy of Nitrogen is more than oxygen.

Ans. Nitrogen is a stable element due to its half-filled p-sub shell. Due to this reason, it has more ionization energy than oxygen.



ELECTRONEGATIVITY:

“Electronegativity is the tendency of an atom to attract shared pair of electrons in a chemical bond.”

- Electron gaining tendency of an element is also called electronegativity.
- Non-metals are electronegative because they have a tendency to accept electron.
- Electronegativity is typically measured on a scale called the Pauling scale, named after the American Quantum Chemist Linus Pauling.
- On the Pauling scale, the electronegativity values range from 0 (least electronegative) to 4 (most electronegative).
- The most electronegative element is Fluorine (F) having E.N value 4.
- The least electronegative element in Cesium (Cs) having E.N value 0.7.
- Electronegativity is inversely proportional to the number of shells and directly proportional to the magnitude of nuclear charge.

General group trend:

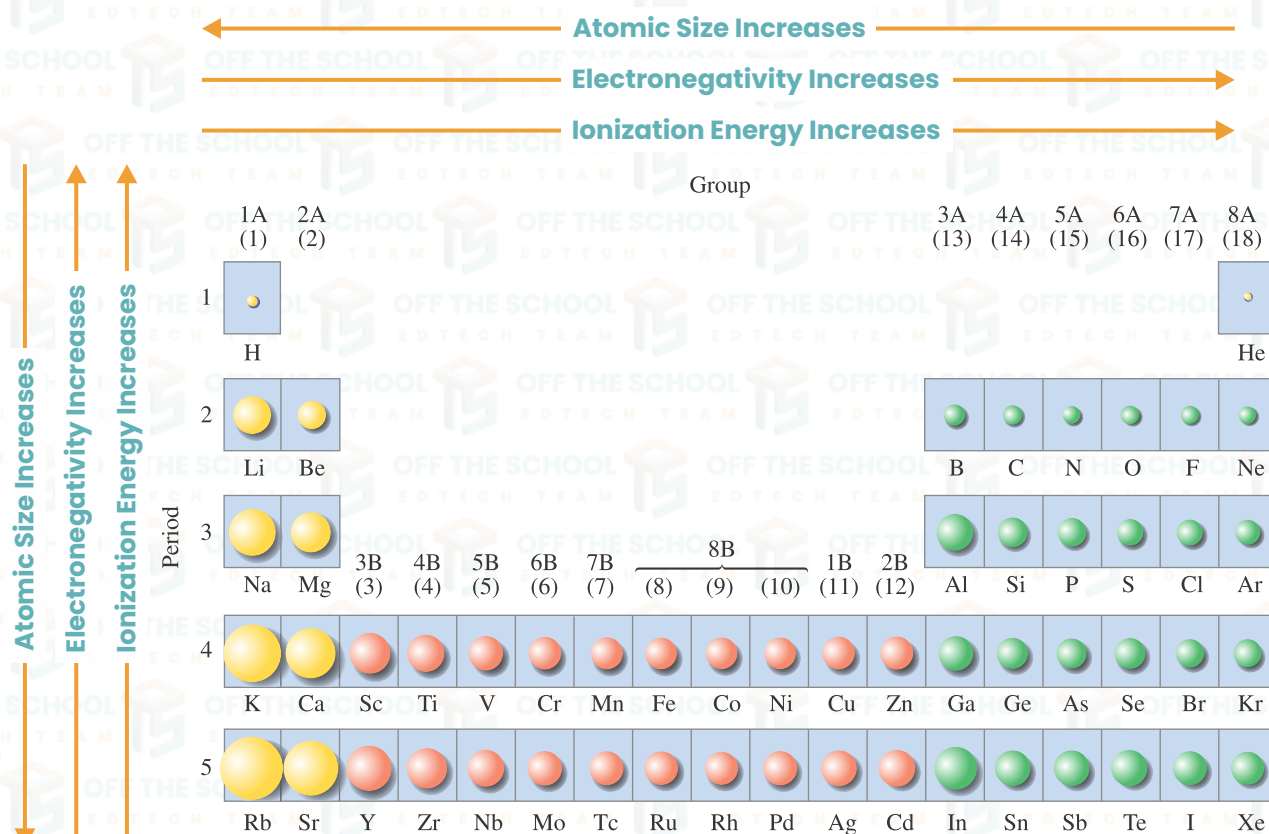
Electronegativity generally decreases as you move down a group. This is because atoms lower in a group have more electron shells, which shield the outermost electrons from the nucleus's positive charge. The increased distance between the outermost electrons and the nucleus reduces the atom's ability to attract electrons, making it less electronegative.

Exception:

Why does the electronegativity (E.N) of group III elements exhibit an irregular trend, initially decreasing from Boron (B) to Aluminum (Al) and then increasing from gallium (Ga) to thallium (Tl)?

Reason:

The irregular increase in electronegativity can be attributed to the poor shielding effect of the electrons in the d-orbital (for the initial decrease) and f-orbitals (for the subsequent increase), respectively.



Highlighting the uniqueness of beryllium

1. Low Atomic Radius: Beryllium has the smallest atomic radius among the alkaline earth metals. Its small size is due to its high effective nuclear charge, which attracts its electrons closely to the nucleus.

2. High Melting Point: Beryllium has the highest melting point among the alkaline earth metals. This property is a result of its strong metallic bonding due to its small size and high nuclear charge.

3. Low Reactivity: Unlike other alkaline earth metals, beryllium is relatively unreactive. It forms a protective oxide layer on its surface, preventing further reaction with air or water. This oxide layer makes it resistant to corrosion.

4. Toxicity: Beryllium and its compounds are highly toxic when inhaled or ingested. This toxicity is a unique and significant characteristic not shared by other group members.

5. Low Density: Beryllium has a lower density as compared to the other alkaline earth metals. Its low density contributes to its lightweight properties.

6. Absence of Diatomic Molecule: Unlike its group members such as magnesium (Mg), calcium (Ca), and strontium (Sr), beryllium does not form stable diatomic molecules in the gas phase. For example, there is no Be_2 molecule.

7. No Alkaline Hydroxide Formation: Beryllium does not readily form alkaline hydroxides like the other alkaline earth metals. While others react with water to form alkaline solutions, beryllium reacts with water sparingly.

8. High Ionization Energy: Beryllium has a higher ionization energy as compared to other group members, making it more challenging to remove electrons from its outermost shell.

These unique characteristics make beryllium stand out within its group and distinguish it from the other alkaline earth metals.

INDUSTRIAL PREPARATION OF SODIUM HYDROXIDE:

Commercial names:

- Caustic soda
- Lye
- Sodium hydrate
- White caustic

CASTNER-KELLNER PROCESS:

Principle:

In castner-kellner method NaOH is prepared by the electrolysis of aqueous solution of NaCl (Brine).

Concentration of brine:

25 % mass/mass; i.e. 25 gm of NaCl is dissolved in 75 gm of water.

CASTNER-KELLNER CELL'S CONSTRUCTION:

Upper chamber

- It is a rectangular tank of steel.
- Inside of tank is lined with "ebonite".
- Anode is made of titanium and dipped in Brine.

Lower chamber

It mainly consists on a pipe in which hydrargyrum is present. Flowing layer of mercury (Hg) at the bottom of tank serves as cathode.



- Mercury pump is fixed for flowing of mercury.
- The two chambers are connected through Denuder.

DETAILS OF PROCESS:

Ionization of NaCl:

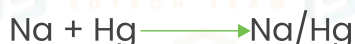


When electric current is passed through brine, +ve and -ve ions migrate towards their respective electrodes. Na^+ ions are discharged at mercury cathode. The sodium deposited at mercury forms Sodium Amalgam (Na/Hg). Chlorine produced at the anode is removed from the top of the cell.

Reaction at cathode



Na forms amalgam



✓ Na^+ ions are discharged in preference to H^+ ions due to high over voltage:

The reduction potential of sodium ions (Na^+) is lower (more negative) than that of hydrogen ions (H^+). In other words, sodium ions are easier to reduce than hydrogen ions. The reduction potential of a species indicates the tendency of that species to gain electrons and be reduced during an electrochemical reaction.

Na^+/Na : E.P./R.P. = -2.71 volt

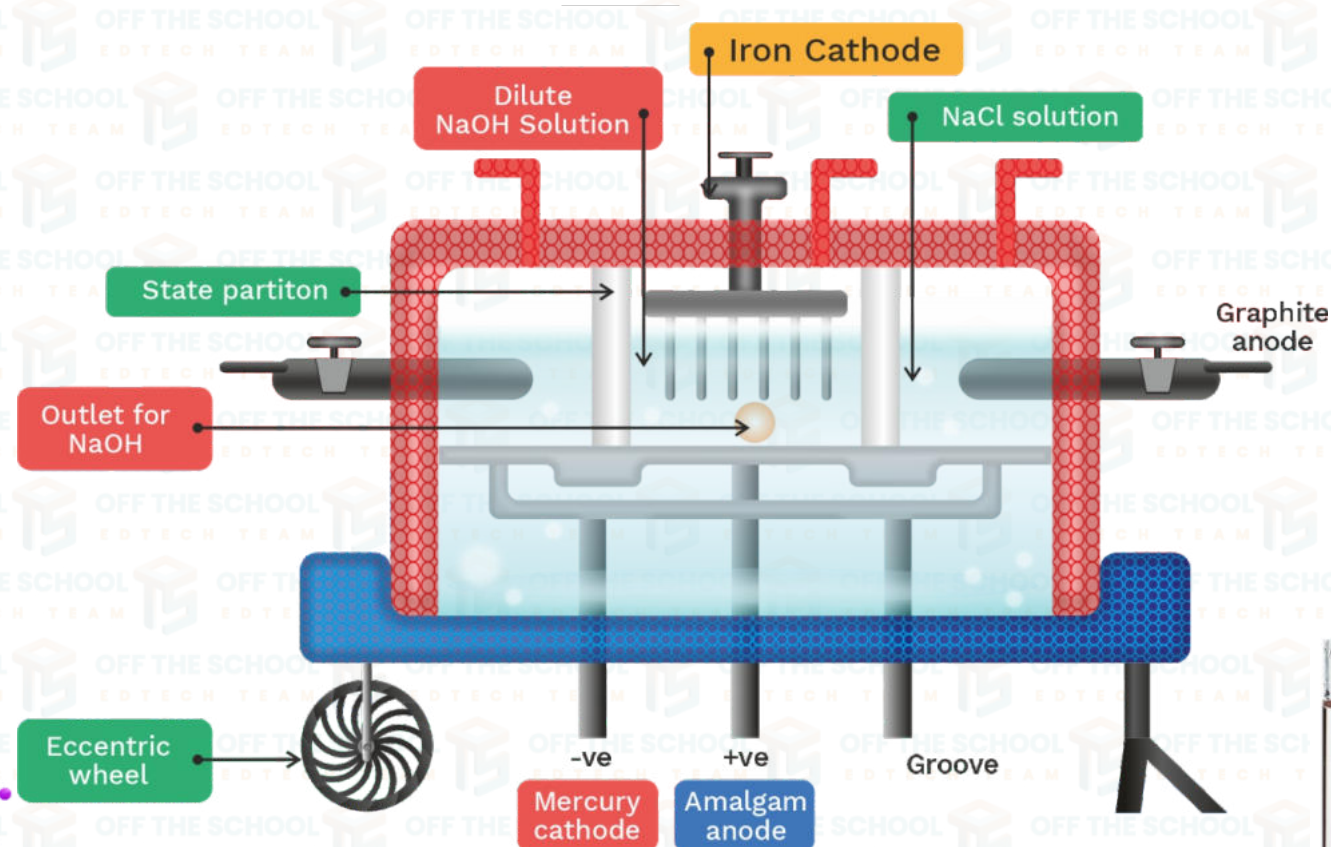
H^+/H : E.P./R.P. = 0.00 volt

Reaction at anode:



Formation of NaOH

Amalgam moves to another chamber called "denuder", where it is treated with water to produce NaOH which is in liquid state. Solid NaOH is obtained by the evaporation of this solution.



INDUSTRIAL PREPARATION OF SULFURIC ACID H_2SO_4 (By contact process)

Sulfuric acid, known as "Oil of vitriol" is a highly corrosive and powerful dehydrating agent widely used in industries, laboratories, and chemical processes. Yes, sulfuric acid was obtained through the distillation of green vitriol in the past. The process was called the Leblanc process, and it was developed in the 17th century.

The Leblanc process involved the following steps:

1. Green vitriol (Iron (II) sulfate) was heated in a furnace, producing sulfur dioxide and sulfur trioxide.
2. The sulfur dioxide and sulfur trioxide were then reacted with water to produce sulfuric acid.
3. The sulfuric acid was then concentrated by boiling it off.

Step 1: Preparation of sulphur dioxide:

SO_2 is prepared by burning sulphur in the presence of excess air so that the product combines with oxygen which is helpful for the next stage.

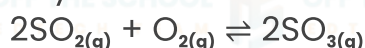


SO_2 is also produced by roasting Iron pyrite (FeS)



Step 2: Preparation of sulphur trioxide:

Sulphur trioxide is formed when sulphur dioxide reacts with oxygen in a ratio of 1 : 1 in the presence of V_2O_5 as a catalyst. This reaction is reversible in nature.



Conditions for Forward reaction

- Low Temperature of about 400°C – 450°C because reaction is Exothermic.
- A pressure of 1–2 atm.
- Take excess O_2 .

Step 3: Absorption of SO_3 :

The sulphur trioxide formed is first made to react with concentrated sulphuric acid. Sulphur trioxide cannot be dissolved in water directly as it leads to the formation of fog. The product obtained after this reaction is known as oleum.

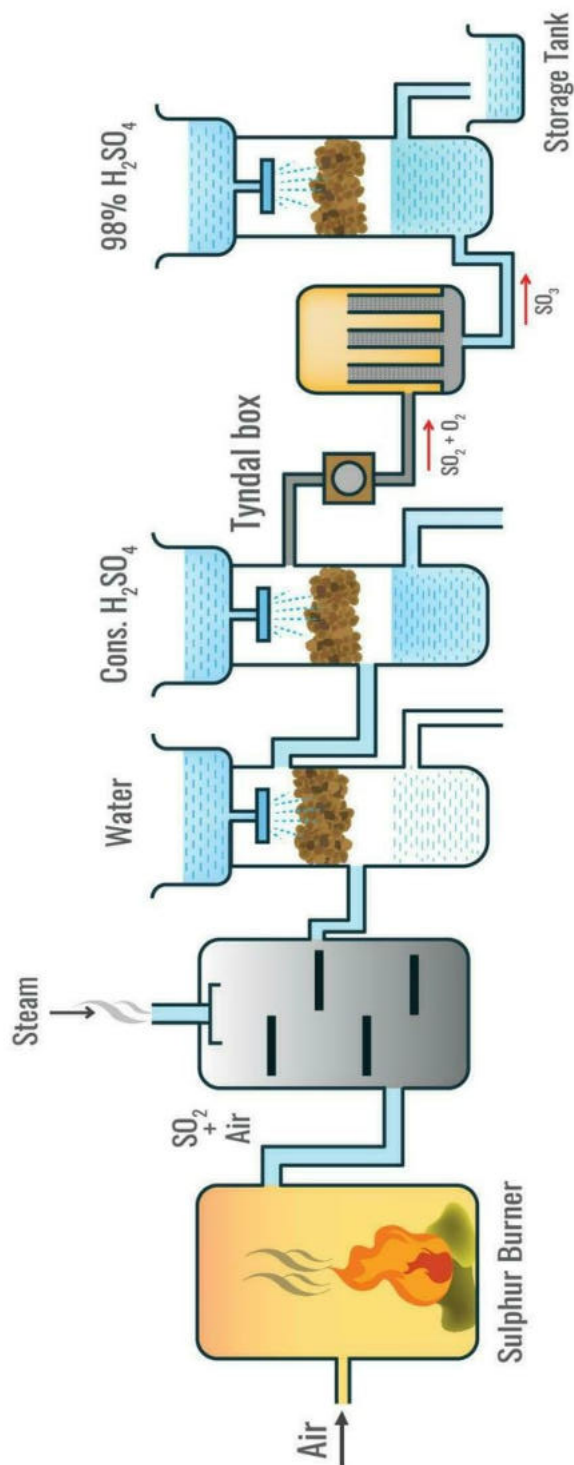


Step 4: Dilution of Oleum:

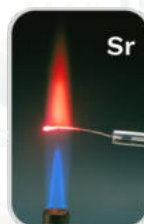
The oleum obtained is then dissolved in water to obtain concentrated sulphuric acid.



Manufacture of H_2SO_4 By Industrial Process



Li	Crimson red
Na	Golden Yellow
K	Violet
Rb	Red Violet
Cs	Blue



The diagonal relationship:

The similarity in properties observed between two elements placed diagonally from left to right in two adjacent periods and two adjacent groups in the periodic table is known as the diagonal relationship.

	1A (1)	2A (2)	3A (13)	4A (14)
2	Li	Be	B	
3		Mg	Al	Si

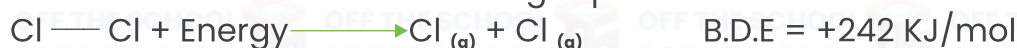
REASONS:

1. Why do elements in the same group of the periodic table often exhibit similar chemical properties?
2. When moving down a group of representative elements, why does the atomic radius generally increase?
3. Explain why the ionization energy tends to decrease when moving down a group of representative elements.
4. Discuss the trend in the reactivity of alkali metals (Group 1) as you go down the group. Why do they become more reactive?
5. Contrast the properties of halogens (Group 17) with noble gases (Group 18). What makes them different, and how does their position in the periodic table explain these differences?
6. In Group 14, why does carbon (C) differ significantly in its properties from silicon (Si), germanium (Ge), and tin (Sn)? What sets carbon apart?

halogen	molecule	structure	model	$d(X-X)$ / pm (gas phase)	$d(X-X)$ / pm (solid phase)
fluorine	F ₂			143	149
chlorine	Cl ₂			199	198
bromine	Br ₂			228	227
iodine	I ₂			266	272
astatine	At ₂				

Bond Enthalpy:

Bond enthalpy, also known as bond energy, is the amount of energy required to break one mole of a chemical bond in the gas phase.



Bond enthalpies of halogens:

- Bond enthalpies of halogens decrease with increasing bond length due to the weaker overlap of atomic orbitals.
- Fluorine is an exception to this trend due to the large repulsion between the non bonding electrons of small sized atoms in the fluorine molecule. Increasing order of Bond enthalpy of halogens are: $\text{Cl} > \text{Br} > \text{F} > \text{I}$

Acidity of hydrogen halides & Bond enthalpy in hydrogen halides:

Hydrogen halides (HX) are strong mineral acids and their acidity follows an increasing trend from HF to HI. ($\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$).

Acidity of hydrogen halides increases with decreasing bond energy because it is easier to donate a proton from a weak bond. ($\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$)

Short key points:

- Bond enthalpies of halogens decrease from chlorine to iodine (except fluorine).
- Acidity of hydrogen halides increases from HF to HI.
- Bond enthalpies of hydrogen halides decreases from HF to HI.

Oxidizing Power of Halogens:

The oxidizing power of a halogen is its ability to oxidize other elements.

The oxidizing power of halogens decreases as you go down the group. This is because the bond length between the halogen and its electron increases as you go down the group. This makes it easier for the halogen to lose its electron and therefore have a lower oxidizing power.

Fluorine have a highest oxidizing power why?

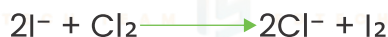
Fluorine has the highest oxidizing power because of the following reasons:

- High electronegativity
- Small size
- Weak bond enthalpy

Reducing Power of Halogens:

The reducing power of a halogen is its ability to be reduced by other elements.

The reducing power of halogens increases as you go down the group. This is because the bond length between the halogen and its electron increases as you go down the group. This makes it easier for the halogen to gain an electron and therefore have a higher reducing power.

Example:

In this reaction, chlorine is the oxidizing agent and iodine is the reducing agent. Chlorine is more electronegative than iodine, so it is able to attract electrons from iodine. This causes the iodine to be oxidized to iodine molecules. The chlorine is reduced to chloride ions.

Here is a breakdown of the oxidation and reduction half-reactions:

Oxidation half-reaction:**Reduction half-reaction:**

Halogen	Oxidizing power	Reducing power
Fluorine (F)	Highest	Lowest
Chlorine (Cl)	High	Medium
Bromine (Br)	Medium	High
Iodine (I)	Low	Highest

Auto-oxidation & reduction of Chlorine:

Auto-oxidation of chlorine is a process in which chlorine reacts with waters to produce hydrochloric acid and hypochlorous acid.

The following is an example of the auto-oxidation of chlorine:

